

$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \quad t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$	• $z > z \text{ or } t > t_{(n-1, \alpha)}$ • $z < -z_\alpha \text{ or } t < -t_{(n-1, \alpha)}$ • $\frac{Z_\alpha}{2} < z < -\frac{Z_\alpha}{2} \text{ or } t_{(n-1, \frac{\alpha}{2})} < t < -t_{(n-1, \frac{\alpha}{2})}$
$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$	$df = n_1 + n_2 - 2$ $t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left[\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1}\right] + \left[\frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}\right]}$
$t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}} \quad s_d = \sqrt{\frac{n \sum d_i^2 - (\sum d_i)^2}{n(n-1)}}$ $z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$	$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{p_1 q_1 / n_1 + p_2 q_2 / n_2}}$ $z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \quad \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$
$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$	$r_{pb} = \frac{M_1 - M_0}{s_n} \sqrt{\frac{n_1 n_0}{n^2}}, \quad s_n = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$
$z = \frac{\sqrt{n-3}}{2} \left[\ln \left(\frac{1+r}{1-r} \right) - \ln \left(\frac{1+\rho_0}{1-\rho_0} \right) \right] = \frac{\sqrt{n-3}}{2} \ln \left[\frac{(1+r)(1-\rho_0)}{(1-r)(1+\rho_0)} \right]$ $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	$Z_r - Z_\alpha \frac{1}{\sqrt{n-3}} < Z < Z_r + Z_\alpha \frac{1}{\sqrt{n-3}} \quad Z_r = 0.5 \ln \left(\frac{1+r}{1-r} \right)$ $r = \frac{e^{2z'} - 1}{e^{2z'} + 1}$
$Y = a + bX \quad X = a + bY$ • $Y - \bar{Y} = b_{yx}(X - \bar{X})$ where $b_{yx} = \frac{N \sum XY - \sum X \sum Y}{N \sum X^2 - (\sum X)^2}$ • $X - \bar{X} = b_{xy}(Y - \bar{Y})$ where $b_{xy} = \frac{N \sum XY - \sum X \sum Y}{N \sum Y^2 - (\sum Y)^2}$	$r = \sqrt{b_{xy} \cdot b_{yx}} \quad \frac{b_{xy} + b_{yx}}{2} \geq r$ $s_{y \cdot x}^2 = \frac{1}{n-2} \sum e_i^2 = \frac{1}{n-2} \sum (y_i - \hat{y}_i)^2$
$b_1 - t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}}, \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$ $s^2 = \frac{S_{yy} - b_1 S_{xy}}{n-2} \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ $t = \frac{b_1 - \beta_{10}}{s / \sqrt{S_{xx}}}$	$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$ $b_1 = \frac{(\sum X_2^2)(\sum X_1 Y) - (\sum X_1 X_2)(\sum X_2 Y)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1 X_2)^2}$ $b_2 = \frac{(\sum X_1^2)(\sum X_2 Y) - (\sum X_1 X_2)(\sum X_1 Y)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1 X_2)^2} \quad R^2 = \frac{b_1 \sum X_1 Y - b_2 \sum X_2 Y}{\sum Y^2}$